

Smart Math Platform: Adapting to Every Primary School Student's Level

Module: Personalized Learning Content (AI-Powered Adaptive Curriculum)

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A Brief Description of the Module: This project aims to improve elementary school students' mathematics learning experiences through an AI-powered personalized content recommendation system. The system will analyze each student's knowledge level, learning pace, and individual needs to dynamically suggest the most suitable topics, questions, exercises, and activities. Through this personalized approach, students will have the opportunity to learn at their own pace, making the learning process more efficient and engaging. In this way, the system will not only support academic achievement but also enhance students' motivation, self-confidence, and positive attitudes toward mathematics. The project ultimately aims to create a more inclusive, equitable, and effective learning environment powered by artificial intelligence.

The Purpose: The main purpose of the AI-Powered Adaptive Curriculum platform is to fundamentally shift the mathematics learning paradigm for primary school students from a standardized, one-size-fits-all model to a truly individualized and effective experience.

Scope: The system's functionalities will include:

- **Initial Assessment:** Conducting a placement test to establish a baseline proficiency level for each student.
- **Adaptive Content Delivery:** Utilizing the AI engine to generate, select, and sequence personalized math exercises and educational resources
- **Real-Time Performance Tracking:** Logging and analyzing student data (response time, error type, completion rate) for continuous adaptation.

- **Feedback and Gamification:** Providing instant, constructive feedback and using motivational elements (badges, points, level-ups) appropriate for primary school children.
- **Reporting Tools:** Generating comprehensive reports and dashboards for students, teachers, and parents.

1. Identify System and Related Actors

1.1 Understand the System and Its Requirements: To achieve its objectives, the system must execute the following fundamental tasks, which constitute the three-stage process at the heart of the adaptive platform:

Stage	Function	Description
Stage 1: Diagnosis	Data Collection and Analysis	The system continuously records and streams student metrics in real-time to the AI engine. Metrics include response speed, answer accuracy, and specific error types (e.g. mistakes in carrying over during addition).
Stage 2: Adaptation	Learning Path Management	Based on the analyzed data, the AI engine dynamically selects the student's next content piece (exercise or topic review). This constitutes the instantaneous personalization of the curriculum.
Stage 3: Presentation and Feedback	Content Delivery	The system presents the selected content in a format suitable for the student's level and learning style and provides immediate, constructive feedback after every interaction.

Users:

- 1- **Student** : The primary user who consumes adaptive content, solves exercises, receives feedback, and whose proficiency level is assessed and adjusted.
- 2- **Teacher**: The authorized user who monitors student performance, analyzes reports, oversees the adaptive path, and can manually intervene or assign specific tasks.
- 3- **Parent/Guardian**: The external user who views the child's progress summaries and development reports, but does not directly influence content delivery.
- 4- **System Administrator / Curriculum Manager**: Personnel responsible for uploading curriculum content, managing overall system settings, and administering user accounts.

1.2 Identify Actors:

Users:

1. The Student Workflow

The student's journey is defined by the AI-controlled adaptive loop:

1. **Login and Initial Assessment**: The Student securely logs into the system. First-time users are directed to a pre-assessment test to establish their baseline proficiency level.
2. **Content Consumption**: The Student begins working on the first content piece (lesson, video, or exercise) assigned by the AI Adaptive Engine.
3. **Practice and Response**: The Student solves the assigned exercises. The system provides instant feedback (correct/incorrect) and assigns points or badges.
4. **Adaptive Redirection**: If the Student makes errors or shows low confidence, the system automatically returns to an easier, foundational exercise or an essential review topic. If the Student succeeds, the system increases the content's difficulty and moves them forward on their personalized path.

2. The Teacher Workflow

The teacher uses the system as a management and diagnostic tool:

1. **Login and Reporting**: The Teacher securely logs in and views the classroom's general performance dashboard (e.g., average mastery scores, time spent on task).

2. **Analytical Review:** The Teacher reviews the detailed analytical reports generated by the AI, which highlight specific concepts (e.g., fractions, measurement) where the class or individual students are struggling.
3. **Assignment:** Based on data analysis, the Teacher can assign a special, supplementary group task or homework to the entire class or a select group of students, focusing on addressing identified shortcomings.

3. The Parent Workflow

The Parent role is typically limited to monitoring and tracking the student's progress and achievements, without academic intervention in the curriculum.

1. **Secure Login and Access:** The Parent/Guardian securely logs into the system using their dedicated account linked to the student's profile.
2. **Summary Dashboard View:** The Parent views a Summary Dashboard that displays their child's overall math proficiency, recently completed tasks, and current motivational/gamification scores (e.g., badges earned, points).
3. **Detailed Report Review:** The Parent can access and review periodic progress reports (weekly/monthly) generated by the AI. These reports highlight specific topics where the child excelled or struggled, and detail the overall learning growth.
4. **Communication Setup:** The Parent utilizes a tool within the system (e.g. integrated email or internal messaging) to easily initiate communication with the child's Teacher.

4. The System Administrator / Curriculum Manager Workflow

The Administrator role is responsible for maintaining the system's infrastructure, content integrity, and user accounts.

1. **Administrative Access:** The Administrator logs into the backend administration panel with high-level privileges.
2. **Content Upload and Tagging:** The Administrator uploads new math lesson materials to the Content Database and ensures they are properly tagged with curriculum standards, difficulty levels, and relevant keywords for the AI engine.
3. **User and Authorization Management:** The Administrator manages the creation, updating, and deletion of Teacher and Student accounts, and sets access permissions (e.g. which Teacher can access which class roster).
4. **System Performance Monitoring:** The Administrator monitors the platform's overall health and performance. They specifically verify that the AI Adaptive Engine and Content Database function correctly under high user load

5. **Data and Security Management:** The Administrator initiates regular data backup processes and verifies compliance with encryption protocols to secure sensitive student data (in support of NFRs regarding security).

External System Actors

These are non-human, critical software or data components essential for the platform's core adaptive functionality.

- **AI Adaptive Engine:** The core computational component that processes real-time student data (speed, error type, mastery level) and uses algorithms to dynamically calculate and determine the optimal next piece of content or difficulty level.
- **Content Database:** The repository responsible for securely storing and efficiently serving all educational assets, e.g. including thousands of categorized math exercises, interactive visuals, and video explanations, which the AI engine selects from.
- **Authentication Service:** A dedicated service responsible for securely verifying user credentials, managing session tokens, and ensuring that only authorized users (students, teachers, parents, system administrator) can access their respective system functions.

1.3.1 Functional Requirements (What the system must do):

1. The system shall allow all Human Users (Students, Teachers, Parents, and Administrators) to create accounts and log in securely.
2. The system shall enforce role-based access control, ensuring that each user can only view and interact with functionalities relevant to their role.
3. The system shall enable Administrators to manually create and manage accounts for Teachers and Students.
4. The system shall provide a pre-assessment test to determine the student's current knowledge and skill level.
5. The system shall record the student's response time, accuracy, and error type in real-time during exercises.
6. The system shall create a dynamic and personalized learning path for the student based on collected data.
7. The system shall automatically offer supplementary content that reverts to relevant foundational concepts when a student makes repeated errors.
8. The system shall systematically increase the difficulty level, directing the student to the next advanced topic following successful performance.

9. The system shall provide instant and explanatory constructive feedback for every task solved by the students.
10. The system shall award badges and points for achievements (completed units, high scores) to boost motivation.
11. The system shall allow Teachers to view detailed learning analytics reports showing class or individual student performance.
12. The system shall grant Teachers the authority to assign custom tasks and homework to students.
13. The system shall provide an access panel for Parents to view their child's progress summaries and development reports.
14. The system shall allow Administrators to upload and tag new curriculum materials compatible with the curriculum.
15. The system shall send automatic reminders (notifications) to students who have not shown progress within a defined time frame.
16. The system shall create an archive for students to review exercises they got wrong and right in the future.
17. The system shall present a mastery test with random questions at the end of the unit to confirm the student has truly grasped the concept.
18. The system shall provide all users with the ability to easily reset or change their passwords within their account settings.
19. The system shall allow Teachers to view the number of students currently logged in to the system in real-time.
20. The system shall automatically save the student's progress when they exit the platform.
21. The system shall display the user's progress graphically.
22. The system shall give some tips for all questions.

1.3.2 Non-Functional Requirements (How the system should perform)

1. The system shall generate new questions within 5 seconds.
2. The system shall support simultaneous access by at least 500 students.
3. The system shall have a simple, colorful, and child-friendly user interface.
4. The system shall securely store students' personal information.
5. The system shall operate smoothly on desktop computers, tablets, and smartphones.
6. The system's artificial intelligence should not show the same questions over and over again.